

Dr. Dipankar Bhattacharya, Ph.D.

Marie Skłodowska-Curie Postdoctoral Fellow (2025–2027)

Dyson School, Imperial College London, United Kingdom

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Research Profile

Independent robotics researcher developing model-based and learning-enabled robotic systems for healthcare, rehabilitation, and advanced manufacturing. My work integrates soft robotics, cable-driven actuation, deformable-object manipulation, perception, and control, including industry-funded esophageal stent testing on the RoSE platform (BvW Holding, NZD 39,000, 2018). As Marie Skłodowska-Curie Fellow at Imperial College London, I lead CASREx (Grant No. 101205678; EUR 276,187.92, 24 months) on cable-driven exosuits for upper-arm muscle-impairment rehabilitation, building on RoSE (Soft Robotics, 2021), cable wrapping modeling (MMT, 2025; IEEE TCST, 2023), and fabric manipulation (IEEE IROS, 2026; IEEE RA-L, 2025).

Education

- 2016 – 2021 📖 **Ph.D., Mechatronics Engineering** in Department of Mechanical and Mechatronics Engineering, The University of Auckland (UoA), 01/08/2016 – 14/04/2021.
Thesis title: *Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthetic Stent Testing and Food Swallow Investigation*
Developed RoSE for physiologically relevant stent testing and swallow investigation; outputs: 3 journal articles (2 in Soft Robotics, 2 IEEE TIE), 2 conference papers, 1 invited book chapter (Springer, 2019).
- 2011 – 2013 📖 **M.Tech , Systems and Control** in Department of Electrical Engineering, Indian Institute of Technology (IIT), 22/07/2011 – 04/07/2013.
Dissertation title: *Non-Linear Predictive Control Model using Relevance Vector Machine*. GPA: **8.4 / 10.0**
Nonlinear MPC with Relevance Vector Machine, validated on a magnetic-levitation system; 1 journal article (Int. J. Sustainable Energy, 2015).
- 2004 – 2010 📖 **B.Tech , Electronics and Communication Engineering**, North Eastern Regional Institute of Science and Technology (NERIST), 21/07/2004 – 23/06/2010.
Project title: *Design and Synthesis of a Sixth Order Tow Thomas Bi-quad filter*. GPA: **9.1 / 10.0**

Employment History (continued)

Employment History

Employment History (continued)

Supervising two Year 3 MEng (Design Engineering) placement students: **Ben Surtees** and **Lijiangke Liu** (March 2026 – Present). Res

Delivered lecture on **Robotics Simulation and Control using PyDrake** for *DESE71002 Robotics Research Projects*, Dysc

Employment History (continued)

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Teaching Experience

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Teaching Experience (continued)

Teaching Experience (continued)

Teaching Experience (continued)

Teaching Experience (continued)

Teaching Experience (continued)

BECE3304 *Neural Networks and Fuzzy Control*; **BECE2010** *Digital Electronics*; **BECE2016** *Signals and Systems*; **BTEE2002** *Network Analysis*

Teaching Experience (continued)

Research Publications

Journal (Published/Accepted)

- 1 Jin*, B., Kobayashi*, A., Bhattacharya, D., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (2025). Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths [*Equal contribution.]. *IEEE Robotics and Automation Letters*, 1–8. <https://doi.org/10.1109/LRA.2025.3619791>
- 2 Bhattacharya, D., Cheung, T. K., Wang, Y., & Lau, D. (2025). Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped Cable-Driven Parallel Robots. *Mechanism and Machine Theory*, 214, 106092. <https://doi.org/10.1016/j.mechmachtheory.2025.106092>
- 3 Bhattacharya, D., Chan, Y. P., Shang, S., Chan, Y. S., Tan, Y., & Lau, D. (2023). Tri-Space Operational Control of Redundant Multilink and Hybrid Cable-Driven Parallel Robots Using an Iterative-Learning based Reactive Approach. *IEEE Transactions on Control Systems and Technology*. <https://doi.org/10.1109/TCST.2023.3263689>
- 4 Bhattacharya, D., Cheng, L. K., & Xu, W. (2021). Nonlinear Model Predictive Control of a Robotic Soft Esophagus for Endoprosthetic Stent Testing. *IEEE Transactions on Industrial Electronics*. <https://doi.org/10.1109/TIE.2021.3121755>

- 5 Bhattacharya, D., Jesna, S., Cheng, L. K., & Xu, W. (2021). RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing. *Soft Robotics*. <https://doi.org/10.1089/soro.2019.0205>
- 6 Dang, Y., Liu, Y., Hashem, R., Bhattacharya, D., Allen, J., Stommel, M., Cheng, L. K., & Xu, W. (2021). SoGut: A Soft Robotic Gastric Simulator. *Soft Robotics*. <https://doi.org/10.1089/soro.2019.0136>
- 7 Bhattacharya, D., Cheng, L. K., & Xu, W. (2020). Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot. *IEEE Transactions on Industrial Electronics*, 67(6), 4711–4720. <https://doi.org/10.1109/TIE.2019.2928239>
- 8 Bhattacharya, D., Nisha, M. G., & Pillai, G. (2015). Relevance vector-machine-based solar cell model. *International Journal of Sustainable Energy*, 34(10), 685–692. <https://doi.org/10.1080/14786451.2014.885030>
- 9 Bhattacharya*, D., Tang*, K., Hang, X., Tokuda, F., Tien, N., & Kosuge, K. (2026). RTFF: Random-to-Target Fabric Flattening Policy Using Dual-Arm Manipulator [accepted. *Equal contribution]. *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- 10 Dong, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Huang, X., Tang, K., Tien, N. C., & Kosuge, K. (2025). Precise Top-layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks. *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*, 1343–1348.
- 11 Bhattacharya, D., Cheng, L. K., & Xu, W. (2018). Testing and Analysis of Migration Displacement of a Flared Stent Deployed in an Esophageal Swallowing Robot. *25th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2018.8600911>
- 12 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2017). Actuation planning and modeling of a soft swallowing robot. *24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2017.8211476>

Books and Chapters

- 1 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2019). *Artificial intelligence approach to the trajectory generation and dynamics of a soft robotic swallowing simulator*. Springer International Publishing. https://doi.org/10.1007/978-3-319-78452-6_1

In Review/Preparation

- 1 Wang, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (n.d.). Automated Fabric Alignment Using Global-Local Optimization and Point Cloud Registration, *IEEE Transactions on Automation Science and Engineering (T-ASE)* [in review].
- 2 Wang, Y., Chow, E., Bhattacharya, D., & Lau, D. (2026). Dual-Stage Spatial and Continuum Cable-Driven Parallel Robot for Airlifting in Bored Pile Construction *Automation in Construction* [in preparation].

Grants and Scholarships

Grants/Fellowships

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|------|---|
| 2025 |  Recipient, Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship for the project “CASREX” (Grant No. 101205678), hosted at Imperial College London . Funded under HORIZON-MSCA-2024-PF-01-01. Duration: 24 months; value: EUR 276,187.92 . |
| 2023 |  Co-author, Innovation and Technology Fund (ITF) Partnership Research Programme (PRP) grant for the project “Flexible Robotic Palm for Human Comforting Touching Sense.”, hosted at the Chinese University of Hong Kong (CUHK). |

Grants and Scholarships (continued)

- 2022 ■ Co-authored Research Talent Hub for Innovation and Technology Fund (RTH-ITF) grant for the project “Hybrid Cable-Driven with Active Soft Backbone Continuum Robot for Dexterous Manipulation in Service Applications.”
- 2018 ■ Recipient, Dynamic Testing of Implant Stents Using Robotic Soft Esophagus (RoSE), BvW Holding, Cham. Duration: 12 months; value: **NZD 39,000**.
- 2018, 2019 ■ Recipient, Riddet Travel Grant, Riddet Institute Centre of Research Excellence (NZD 3,000).

Scholarships

- 2019–2020 ■ Recipient, Doctor of Philosophy Scholarship, University of Auckland.
- 2016–2019 ■ Recipient, Doctor of Philosophy Scholarship, Riddet Institute Centre of Research Excellence.
- 2011–2013 ■ Recipient, Master of Technology Scholarship, Ministry of Human Resource Development (MHRD), Government of India.

Academic Awards, and Miscellaneous Achievements

Academic Awards

- 15/12/25 ■ Selected, **European Talent Academy 2026** (Imperial College London, TU Munich, Politecnico di Milano).
- 2017 ■ Recipient, **Best Conference paper award**, 5th International Conference on Robotic Intelligence Technology and Applications (RiTA), Daejeon, Korea.
- **Ranked 2nd** in poster presentation in Delivery of Functionality in Complex Food Systems (DoF 2017) Conference.

Competitive Examinations

- 2011 ■ **Percentile: 99.12% (Rank 1,211/1,37,853)** in Graduate Aptitude Test of Engineering (GATE).
- 2010 ■ **Percentile: 96.00%** in Graduate Aptitude Test of Engineering (GATE).
- 2004 ■ **Ranked 31st** in All India Junior mathematics Olympiad.

Professional Activities

- 05/02/26 ■ Invited Speaker, **Dyson Engineering Departmental Seminar**, Imperial College London.
- 29/01/26 ■ Invited Speaker, **Imperial Robotics Postgrad Committee**, Imperial College London. Talk: *Learning-based Fabric Manipulation with Mesh Perception and Visual Servoing*.
- 28/01/26 ■ Delivered lecture on **Robotics Simulation and Control using PyDrake** for DESE71002 Robotics Research Projects, Dyson School of Engineering, Imperial College London.
- 08/08/25 ■ Invited Speaker, **World Robotics Conference (WRC 2025)**, Beijing, China. Talk: *Fabric modeling and perception for garment manufacturing*.
- 03/08/25 ■ Workshop Organizer and Presenter, **IEEE ICMA 2025**, Beijing, China. Talk: *Mesh-based Fabric State Estimation and Manipulation*.
- 29/05/23 ■ Poster Presentation, **IEEE ICRA 2023**, ExCeL London, United Kingdom.
- 2018 ■ Presenter, **BioMouth 2018 Meet**, Auckland, New Zealand.
- 2021 - Current ■ Reviewer of IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA), ASME Journal of Mechanisms and Robotics (JMR).
- 2017 - Current ■ Member of the Institute of Electrical and Electronics Engineers (IEEE).
- 2014 ■ Former member of the Electrical Engineering Society (EES), IIT Roorkee, and NERIST Electronics Society (NES), NERIST.

Academic Awards, and Miscellaneous Achievements (continued)

- 2006  Have an 'Outstanding Certificate' in National Service Scheme (NSS) for contributing in activities like blood donation camps, village cleanliness programmes and children education.

Skills

Languages	 English and Hindi: professional reading, writing, and speaking proficiency.
Research areas	 Soft robotics, cable-driven robots, robotic perception, deformable-object manipulation, digital twin and control, learning-based modeling and control, and robotic systems for healthcare and manufacturing.
Programming	 Python, MATLAB, PyTorch, ROS2, LabVIEW, assembly-level programming (8085, 8086, MASM32, and 8051), Git, and LaTeX.
Machine learning	 Imitation learning, reinforcement learning, sparse identification modelling, nonlinear MPC with learned dynamics, mesh-based deep perception for deformable objects.
Simulation	 NVIDIA Isaac Sim, pydrake, LeRobot, OpenSim
CAD and others	 Onshape, Blender, SOLIDWORKS, Autodesk Inventor, Inkscape, and Lightroom.
Hardware	 NVIDIA Jetson, Raspberry Pi, myRIO, and Arduino.

References

Prof Kazuhiro Kosuge

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